Biology: Cells and Organelles

Cell theory

• All living things are made of one or more cells.

• The cell is the basic unit of life.

• All cells come from pre-existing cells.

Two main cell types

• **Prokaryotic cells (bacteria and archaea):** No nucleus or membrane-bound organelles. DNA is located in a nucleoid region within the cytoplasm. Usually smaller and structurally simpler than eukaryotic cells.

• **Eukaryotic cells (plants, animals, fungi, and protists):** Have a nucleus and membrane-bound organelles. Usually larger and more complex.

Key organelles and cell structures

• **Nucleus:** Holds DNA and directs cell activities.

• **Mitochondria:** Produce ATP through cellular respiration; often called the cell's powerhouse.

• **Ribosomes:** Build proteins.

• **Endoplasmic reticulum (ER):** Rough ER has ribosomes and helps make and process proteins. Smooth ER makes lipids and helps with detoxification.

• **Golgi apparatus:** Modifies, sorts, and packages proteins for transport.

• **Lysosomes:** Break down waste and worn-out cell components; commonly emphasized in animal cells.

• **Cell membrane:** Selectively permeable barrier that controls what enters and leaves the cell.

• **Cytoplasm:** The contents between the cell membrane and nucleus in eukaryotic cells, including the jelly-like cytosol and organelles.

Plant vs. animal cells

• Plant cells have a **cell wall** for rigid support and typically a **large central vacuole** that stores water. Photosynthetic plant cells have **chloroplasts** for photosynthesis. Animal cells lack cell walls and chloroplasts.

• Cell walls are not exclusive to plants: fungi and many prokaryotes also have them.

• Animal cells typically have **centrioles**. Their shapes vary, but they are often rounder or more irregular than plant cells.